

Module 6: Metabolism — Study Questions

1. What is the definition of energy?
2. What is the difference between kinetic energy and potential energy?
3. The First Law of Thermodynamics says energy cannot be created or destroyed. How does this apply to living cells?
4. The Second Law of Thermodynamics says entropy (disorder) is always increasing. How do cells maintain their order?
5. What is entropy?
6. What is metabolism?
7. What is the difference between exergonic and endergonic reactions?
8. What does ATP stand for?
9. Why is ATP considered the "energy currency" of the cell?
10. What is an enzyme?
11. What is a substrate?
12. What is the "active site" of an enzyme?
13. What is "energy of activation"?
14. How do enzymes affect the energy of activation?
15. Name two factors that affect enzymatic rate.
16. What does it mean to "denature" an enzyme?
17. What is the difference between competitive and noncompetitive inhibition?
18. In a redox reaction, what does "oxidation" mean?
19. What is the overall equation for photosynthesis?
20. What is the overall equation for cellular respiration?